Deep Learning – Project Phase

Project Kick-off

April 9th 2026

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Timeline and overview

The project runs from *today* **April 9th 2025** and for two weeks (Weeks 5, 6 and 7)! We **end April, 22th, at 12:30 noon (submission deadline)**

Idea Presentation: We have a **check-back on April 15th** where you present your project plan to everyone to get feedback from us and your peers.

There are three projects:

MBE, MBF, MGI

You will work in groups of 2 students

Each group works in only one project assignment

(cannot change after today)

- Do not work alone - make sure there are teams of two!

Learning objectives

Lecture phase (Weeks 1,2,3,4) – **Done!**

✓ Explain basic concepts and maths about neural nets

✓ Understand and discuss papers in deep learning

✓ Describe characteristics and applications

- Deliverables: Quizes 1 & 2

Assessment:

Quiz 1: 25%
Quiz 2: 25%

Project Phase (weeks 5,6,7) – **Starting April 9th!**

- Propose your own approaches for solving problems
- Implement your own approaches in python
  - Deliverable: Project Report.

Assessment:

Project Report: 50%

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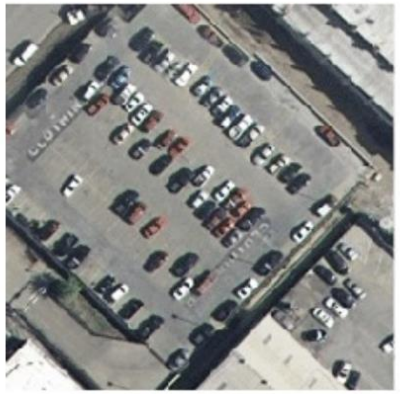
MGI Project - **UC Merced Land Use Dataset**

Tasks:

- image classification(easier),
- multi-label classification (harder)

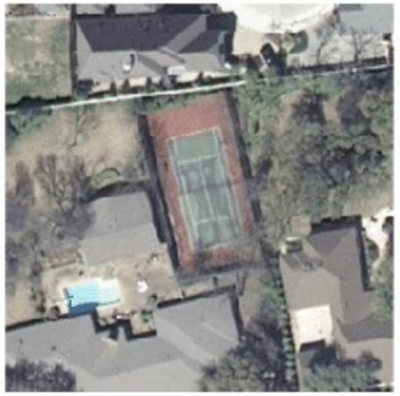


Parking Lot

River

Runway

500 MB in total

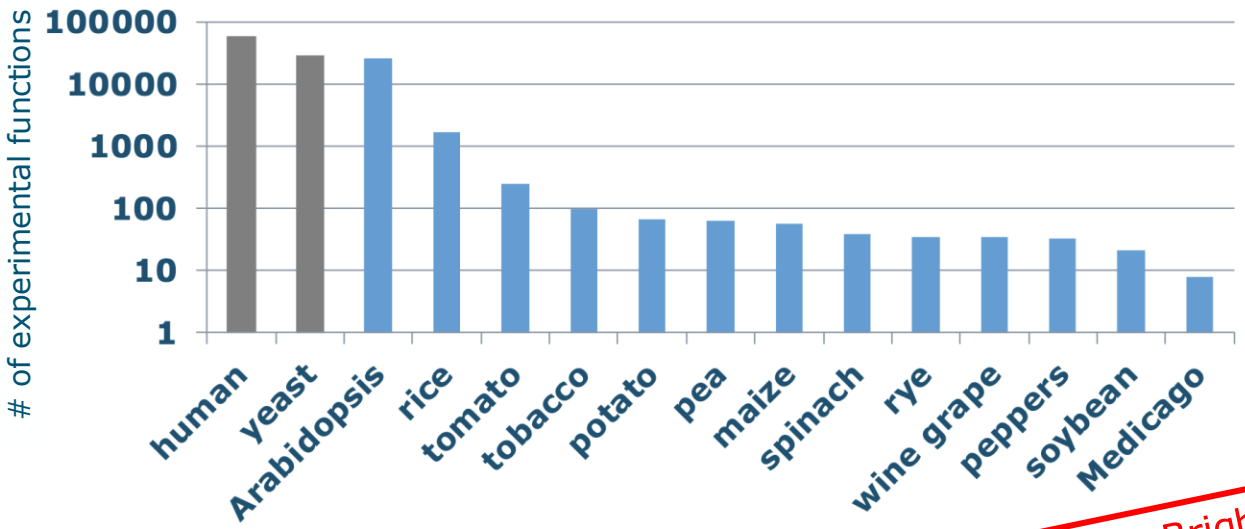


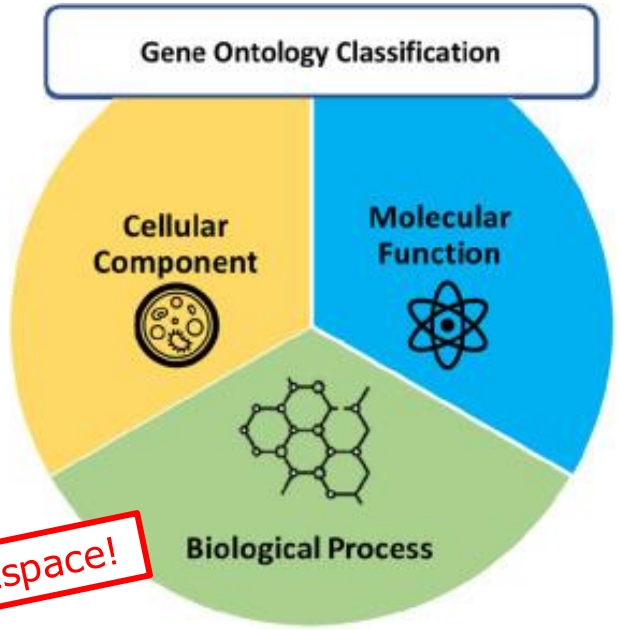
Tennis Court

In the brightspace notebook!

MBF Project – Protein sequence functional annotation

- Tasks: predicting Gene Ontology (GO) terms from protein sequence
  - 6 simulated datasets – increasing level of difficulty
  - 6,784 human protein sequences with 5 different GO terms
  - 14,765 human protein sequences as final test set
- Start with 1D CNN (proof of concept), then finetune transformer





More information on Brightspace!

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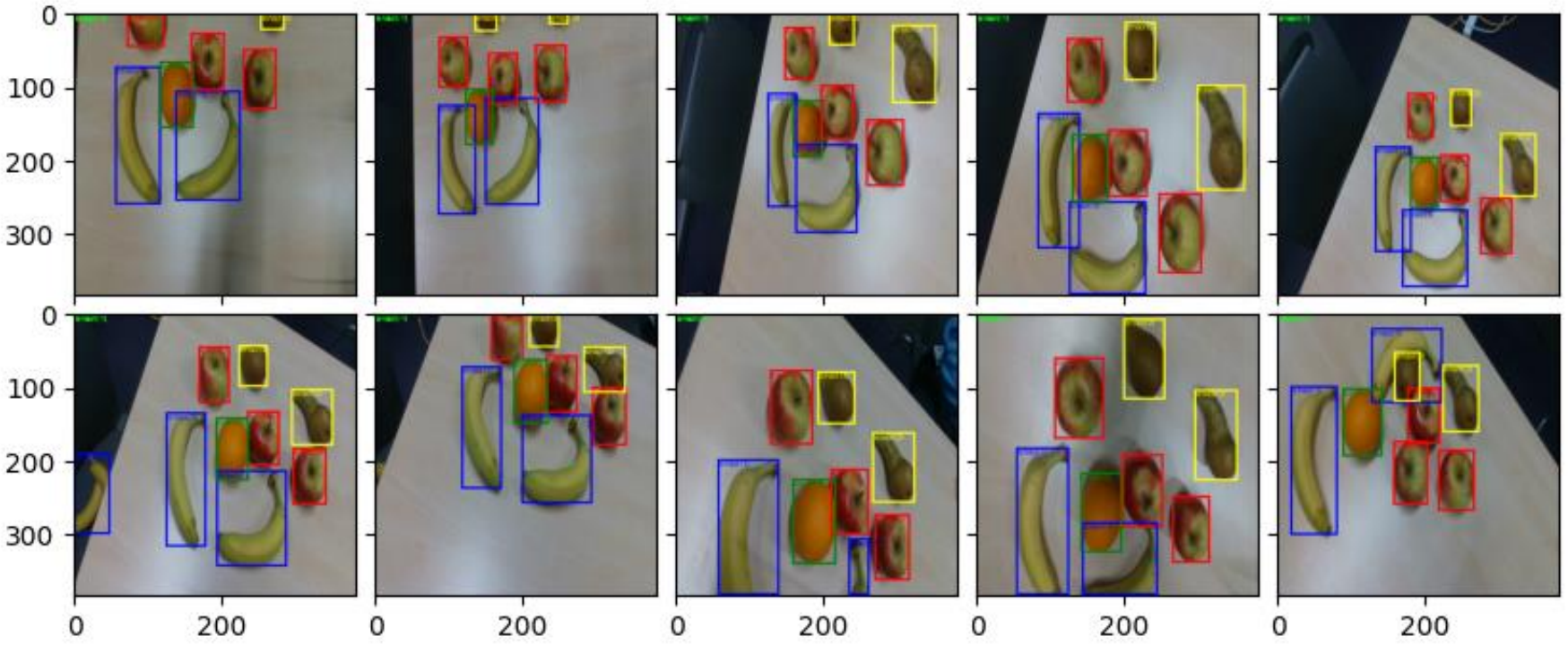
MBE Project – Fruit detection

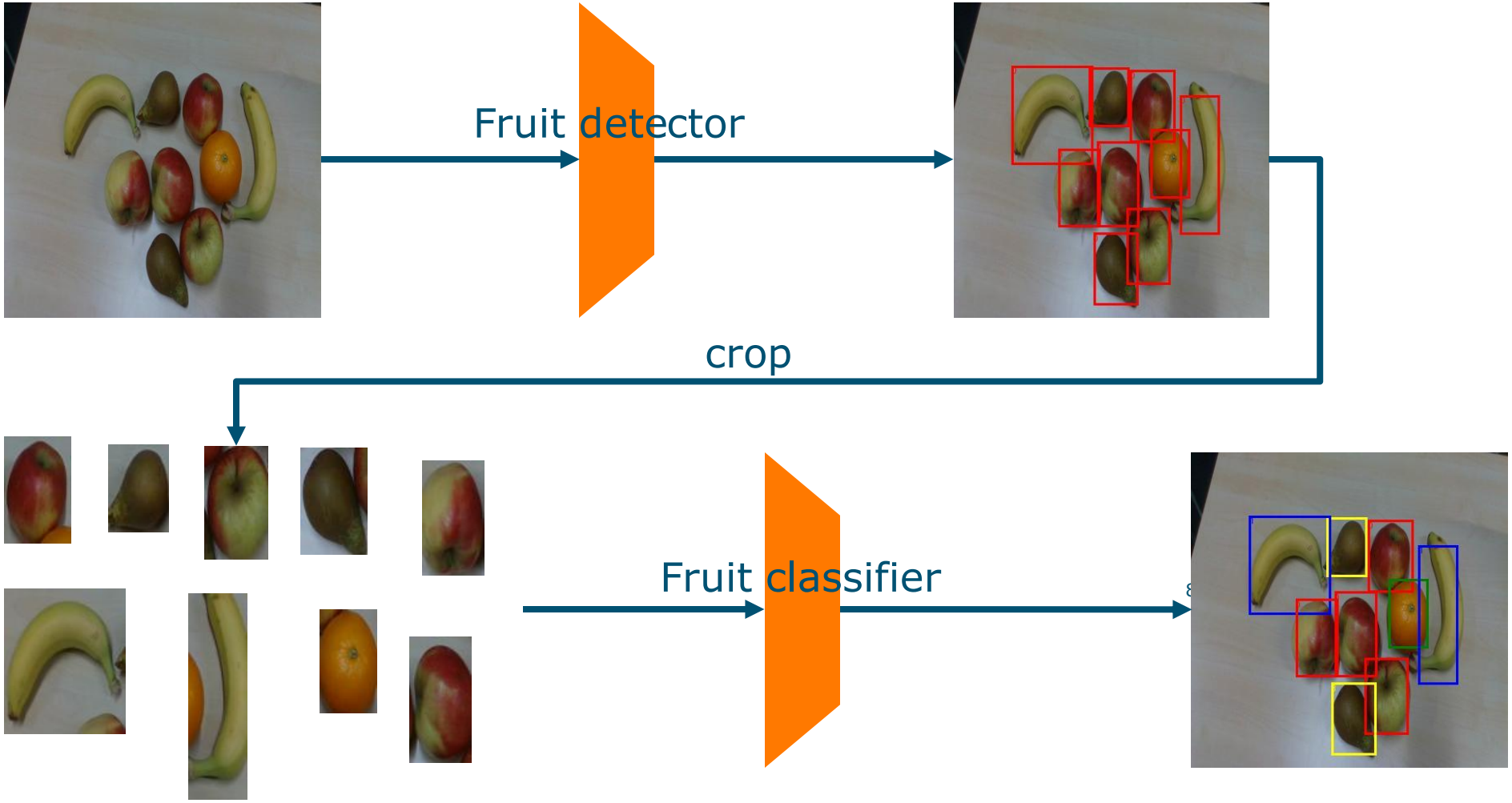
- You'll perform object detection to detect: apples, bananas, oranges and pears



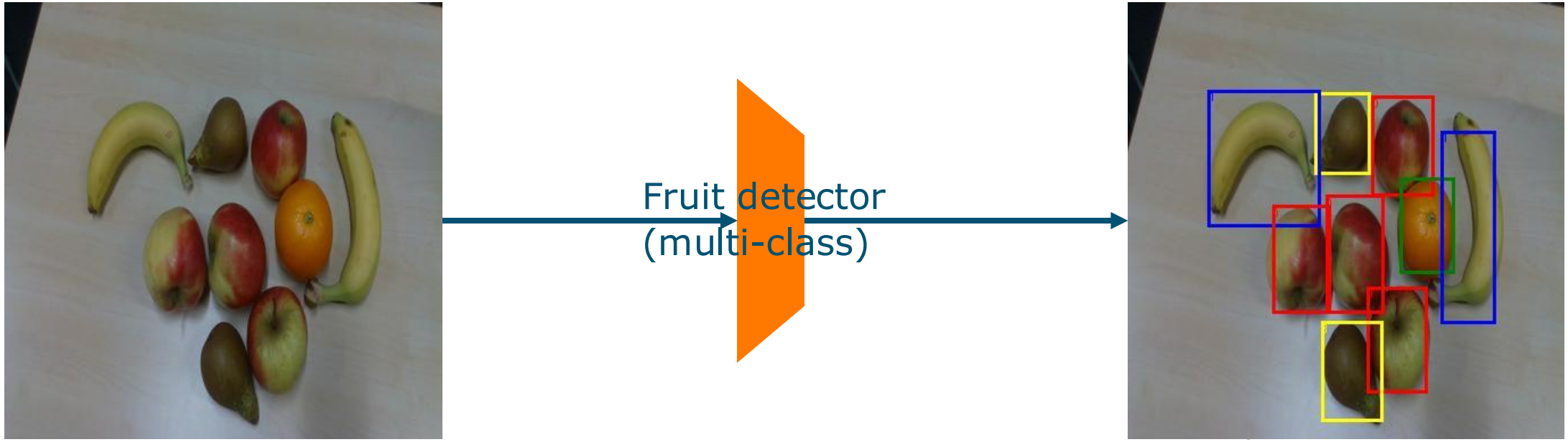
Two methods

- Two-step approach:
  - Single class fruit detector + Fruit classifier
- Single-step approach
  - Multi class fruit detector

Two–step approach



Single–step approach



What we expect from you!

- To implement methods using the knowledge you gained during the course
- To propose solutions for all tasks
- To collaborate within your team!
- Where possible: to compare with state of the art (take numbers from papers, no need to implement their models), but compare wisely on the same sets.
- You don’t have to beat all SOTA ;)
- To be critical about what you did and discuss your methodological choices
- To have FUN ☺

Report structure

1. Brief intro (half page)
2. Methods: explain your approach (2-3 pages)
3. Results and discussion (3-5 pages)
4. Implementation: Link to your code – use a Github or Gitlab repository. Please add also here the shareable link of your code (see next slide)
5. References

Note

- Figures: Refer to figures and tables throughout the text, and embed them within the document close to where they are referenced

What we do/what we don’t

We **do**

- Give you explanations about comprehension of the project
- Come to your rescue if you have software calamities
- Answer to your questions on Teams.

We **don’t**

- Give you programming support, it’s an *exam*
- Tell you if it is right or wrong or if you choose the right strategy
- Read your report beforehand

Groups!

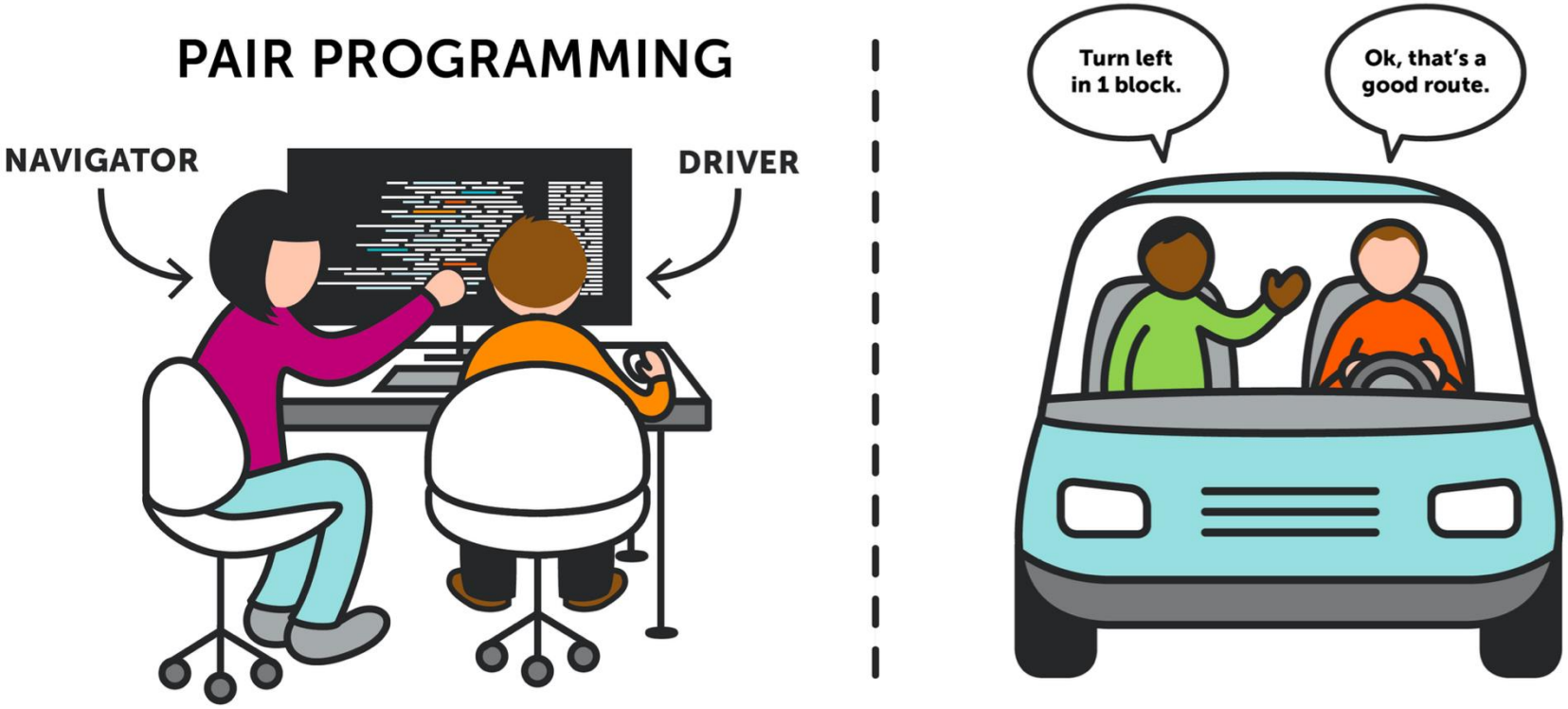
- Is everyone in a group on Brightspace?

Some Technical Tips

1. It is always a good idea to work in Git Repositories like Github or Gitlab. This way you can revert back to previous changes
2. Use logging software like **Weights & Biases to log** and catalogue your model runs
3. Use higher-level packages like **pytorch lightning** to simplify your training, logging, and code
4. You may have to buy additional GPU- resources with Google Colab (we don’t ask you to buy books, so that’s a possible expense) – We may be able to give you access to some of our GPU servers, but then you need to write your code in scripts

Share notebooks and documents

Share notebooks and documents



Submitting your project

You need to submit your project on Wednesday April 22, at 12:30 noon.

You need to submit:

- A report, max 15 pages, including references and figures.

We won't read further.

- Your code, as a python a Github Repository containing scripts or notebooks.

Please add a readme.md to the github repository describing your repository structure. In Notebooks include text blocks, and comments in scripts